

ABHISHEK SATYAM

Senior Software Engineer | React · Next.js · TypeScript | Enterprise Security · RAG / LLM Systems

Bangalore, India | abhishek.satyam96@gmail.com | [+91 9915121582](tel:+919915121582) | linkedin.com/in/abhishek-satyam | abhisheksatyam.com

PROFESSIONAL SUMMARY

Senior Software Engineer with 7 years building production web applications in React, Next.js and TypeScript. Currently at QNu Labs, having built 3 web products from scratch, including QKMS — a live enterprise quantum-safe key management platform — plus their CI/CD and GCP delivery. Previously led a 6-engineer team building a SaaS platform for 50+ enterprise clients across 12+ major releases. Recently designed and shipped a multi-tenant RAG system end to end on Node.js, PostgreSQL and pgvector.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), SQL, HTML5, CSS3

Frontend: React.js, Next.js, Redux Toolkit, React Context, Apollo Client, Tailwind CSS, shadcn/ui, Ant Design, Styled Components, Web Accessibility (WCAG)

Backend & APIs: Node.js, Express.js, REST APIs, GraphQL, OpenAPI, Contract-Driven Integration, NDJSON / SSE Streaming

Databases: PostgreSQL, pgvector, MongoDB — Schema Design, Indexing, Query Optimization

Gen AI / LLM: Retrieval-Augmented Generation (RAG), OpenAI API, Embeddings, Vector Search (HNSW, IVFFlat), Chunking Strategies, Token Streaming, Citation Grounding

Architecture: System Design, Component Architecture, Micro-Frontends, Multi-Tenancy, SSR / SSG / ISR, Caching Strategies

Performance & Monitoring: Core Web Vitals (LCP, CLS, INP), Lighthouse, Lazy Loading, Critical CSS, Image Optimization (WebP / AVIF), Sentry

Testing: Jest, React Testing Library, Integration Testing, Automated Regression in CI

DevOps & Cloud: Docker, Nginx, GitLab CI/CD, GCP, AWS

Build & Tooling: Webpack, Vite, ESLint, Husky, SonarQube

Practices: Feature Flags (LaunchDarkly, Split), A/B Testing, Agile / Scrum

PROJECTS

RAG Knowledge Assistant — TS · Node.js / Express 5 · Prisma · PostgreSQL (Neon) + pgvector · Next.js 16 · OpenAI
rag.abhisheksatyam.com · github.com/AbhishekSatyam96/rag-assistant

- Built a **multi-tenant document Q&A system end to end** — SHA-256 dedupe, recursive chunking, batched embeddings (100/request), and a retrieval path that streams citation-grounded answers and abstains when the corpus cannot support one.
- Designed the vector layer on **PostgreSQL + pgvector**: HNSW index (vector_cosine_ops, m=16, ef_construction=64) tuned per request via transaction-scoped SET LOCAL — chosen over IVFFlat, whose centroids degrade permanently on an incrementally grown table.
- Enforced tenant isolation in raw SQL where the ORM's type safety does not reach, using pgvector 0.8 iterative scan (strict_order) so userId post-filtering cannot silently return fewer than top-k results.
- Streamed tokens over **NDJSON** (fetch + ReadableStream) rather than SSE — EventSource cannot send a Bearer header — and propagated client disconnects to OpenAI via AbortController so abandoned requests stop billing.

WORK EXPERIENCE

Senior Software Engineer, QNu Labs

Mar 2025 – Present

Post-quantum cryptography and quantum-safe key management

- **Built 3 web products from scratch** — QVault, its ground-up successor QKMS, and an internal engineering analytics dashboard — owning repository setup, architecture and toolchain (TypeScript, Tailwind, Ant Design, shadcn/ui, ESLint, Husky) on each, within a 6-person product team.
- Delivered **QKMS (Quantum Key Management System)**, a live enterprise key-management platform — **multi-role authentication** and key profile management — translating cryptographic key operations into workflows enterprise security admins can run safely.
- Designed and shipped an **engineering analytics dashboard on the Jira Cloud REST API** — authored JQL queries, aggregated paginated issue and changelog data, and rendered custom charts on team activity and delivery throughput; **demoed to the CXO team** and adopted for leadership reporting.
- Own **Dockerized GitLab CI/CD pipelines and GCP deployments**, standardizing build, test and release workflows across all 3 products.
- Improved Core Web Vitals across key pages — **reduced LCP from 3.8s to 1.9s**, brought CLS below 0.1, and held Lighthouse performance at 90+.

- Established code-review and unit-test standards (**75%+ coverage**, Jest + RTL) and integrated automated regression testing into CI, reducing recurring production defects and improving release reliability.

Senior Software Engineer, AntWalk

Dec 2019 – Mar 2025

Promoted from Software Engineer, Jun 2020 · *Enterprise learning & capability-building SaaS*

- **Led and mentored a 6-engineer team**, building **2 web applications from scratch** including the core SaaS platform serving **50+ enterprise clients** — owning system design, implementation, testing and deployment across 12+ major releases.
- Architected a library of **40+ reusable React, Next.js and TypeScript components**, cutting feature-development time ~35% and ensuring design consistency across the product.
- Defined and integrated **API contracts for 20+ endpoints** (REST and **GraphQL via Apollo Client**) with backend teams, reducing integration bugs ~25% through contract-driven development and shared OpenAPI documentation.
- Optimized performance via lazy loading, critical CSS extraction and WebP / AVIF image optimization, reducing average page load time ~40% and **improving Lighthouse score from 62 to 88**.
- Increased unit-test coverage from **22% to 72%** (Jest, React Testing Library) and ran 500+ code reviews, establishing engineering standards adopted across the team.
- Built feature-flag (LaunchDarkly, Split) and A/B-testing frameworks across 8+ experiments, enabling data-driven product decisions.

Program Analyst, Cognizant

Jul 2019 – Dec 2019

- Built and maintained frontend modules for an enterprise web app used by 5K+ internal users, contributing to a ~15% reduction in UI-related defects.

EDUCATION

B.Tech, Computer Science & Engineering

2015 – 2019

Lovely Professional University, Phagwara, Punjab

Winner, Techfest Coding Competition, LPU (2019)